COMMUNICATION

Late Cenozoic paleomagnetic record of Duck Hawk Bluffs, Banks Island, Canadian Arctic Archipelago[1]

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Detailed paleomagnetic investigations have been completed on unconsolidated sediments from Duck Hawk Bluffs on Banks Island, in the Canadian Arctic Archipelago, that record some of the oldest late Cenozoic glacial and nonglacial events in Canada. The preglacial Worth Point Formation, the overlying Duck Hawk Bluffs Formation, including marine and glacial deposits laid down during the Banks Glaciation, and the lower part of the interglacial Morgan Bluffs Formation have magnetically reversed directions and therefore are of Matuyama age (>790 ka). Upper Morgan Bluffs Formation organic beds and deposits of the younger Thomsen Glaciation, Cape Collinson Interglaciation, and Amundsen Glaciation are normally magnetized and therefore of Brunhes age (<790 ka). The Brunhes–Matuyama boundary is recorded in the upper portion of the Morgan Bluffs Formation. Its precise position within the interglacial sequence can be identified, since the sediments document the gradual change from reversely inclined directions to normally inclined ones. These results confirm that the preglacial Worth Point Formation is at least Early Pleistocene in age and that the Banks Glaciation (the oldest and strongest continental glaciation recorded in the western Arctic) and a good part of the Morgan Bluffs Formation are of Early Pleistocene age. The study also documents a rare site in Canada where terrestrial sediments record the Brunhes–Matuyama transition and in doing so permits a precise correlation of part of the Banks Island stratigraphy with other key late Tertiary and Early to Middle Pleistocene arctic terrestrial and marine sequences.

Une analyse paléomagnétique détaillée a été complétée sur les sédiments non consolidés des falaises Duck Hawk sur l'ile de Banks dans l'archipel arctique canadien. Ces sédiments témoignent d'événements glaciaires et non glaciaires parmi les plus vieux du Cénozoique supérieur au Canada. Les sédiments préglaciaires de la Formation de Worth Point, les sédiments glaciaires et marins sus-jacents de la Formation de Duck Hawk Bluffs mis en place au cours de la Glaciation de Banks, et les sédiments de la partie basale de la Formation interglaciaire de Morgan Bluffs ont une aimantation inversée et datent donc de l'époque de Matuyama (>790 ka). Les sédiments organiques de la partie supérieure de la Formation de Morgan Bluffs ainsi que les dépôts plus jeunes de la Glaciation de Thomsen, de l'Interglaciaire de Cape Collinson et de la Glaciation d'Amundsen ont une aimantation normale et datent donc de l'époque de Brunhes (<790 ka). La limite entre les époques de Brunhes et de Matuyama est sise dans la partie supérieure de la Formation interglaciaire de Morgan Bluffs et peut être localisée précisément grâce à la transition graduelle des polarités inverses aux polarités normales. Les résultats permettent d'avancer que la Formation préglaciaire de Worth Point date au moins du Pléistocène inférieur et que la Glaciation de Banks (la plus vieille et la plus puissante glaciation continentale reconnue dans l'ouest de l'Arctique) et une bonne partie de la Formation de Morgan Bluffs date également du Pléistocène inférieur. L'étude rend aussi compte d'un des rares sites au Canada où la limite Brunhes–Matuyama peut être localisée dans des sédiments terrestres. Ce faisant, la corrélation de la séquence quaternaire de l'ile de Banks avec d'autres séquences terrestres et marines clés du Pléistocène inférieur et moyen de l'Arctique est donc facilitée.

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Introduction

Sediments on Banks Island in the southwestern Canadian Arctic Archipelago (Fig. 1) record late Tertiary preglacial events as well as three full Quaternary continental glaciations followed or preceded by warm, nonglacial intervals. A full account of these events has been presented in Vincent (1983, 1984, 1989).

Although fossil material is abundant and well preserved, obtaining chronological control on Banks Island is difficult because most sediments are too old for radiocarbon dating. Some success has been obtained in dating Late Pleistocene deposits with the Th–U method (Causse and Vincent 1989), but so far paleomagnetic investigations have been the most useful for obtaining age control for the oldest part of the record and in particular for separating Early Pleistocene from Middle Pleistocene deposits. By convention, the break between the Early and Middle Pleistocene is set at the Brunhes–Matuyama boundary. This reversal has not been directly dated, but here we follow the arguments of Richmond and Fullerton (1986) in accepting an age of 790 ka (Johnson 1982).

Sediments at Duck Hawk Bluffs along the southwestern coast of Banks Island (Fig. 1) provide an excellent record of late Tertiary and Quaternary events. The lithostratigraphy and biostratigraphy of the bluffs have been discussed in Vincent *et al.* (1983) and Matthews *et al.* (1986) and are summarized in Figs. 2 and 3. Tertiary Beaufort Formation sands and gravels are overlain by preglacial Worth Point Formation terrestrial beds, by Banks Glaciation till and marine deposits assigned to the Duck Hawk Bluffs Formation, by interglacial Morgan Bluffs Formation deposits, by marine sediments associated with the Thomsen Glaciation and assigned to the Nelson River Formation, by interglacial Cape Collinson Formation deposits, by Amundsen Glaciation till and marine deposits assigned to the Prince of Wales Formation, and finally by Holocene terrestrial deposits.

Preliminary paleomagnetic investigations (Vincent *et al.*

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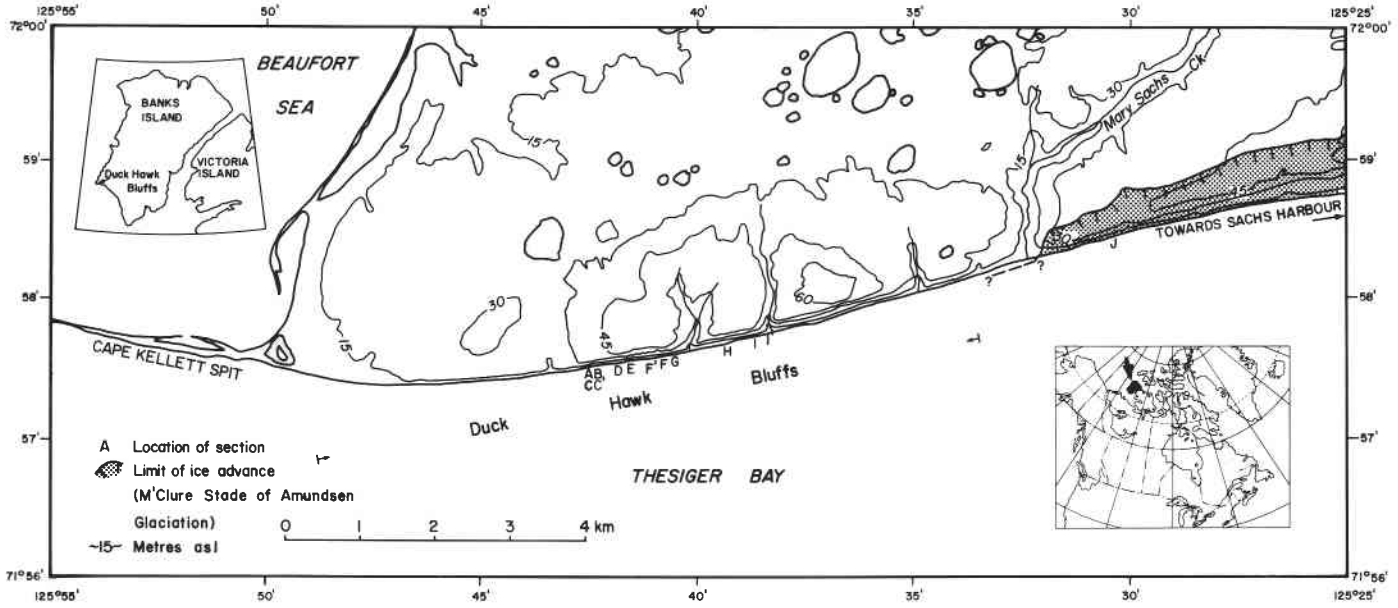


FIG. 1. Location map of study area.

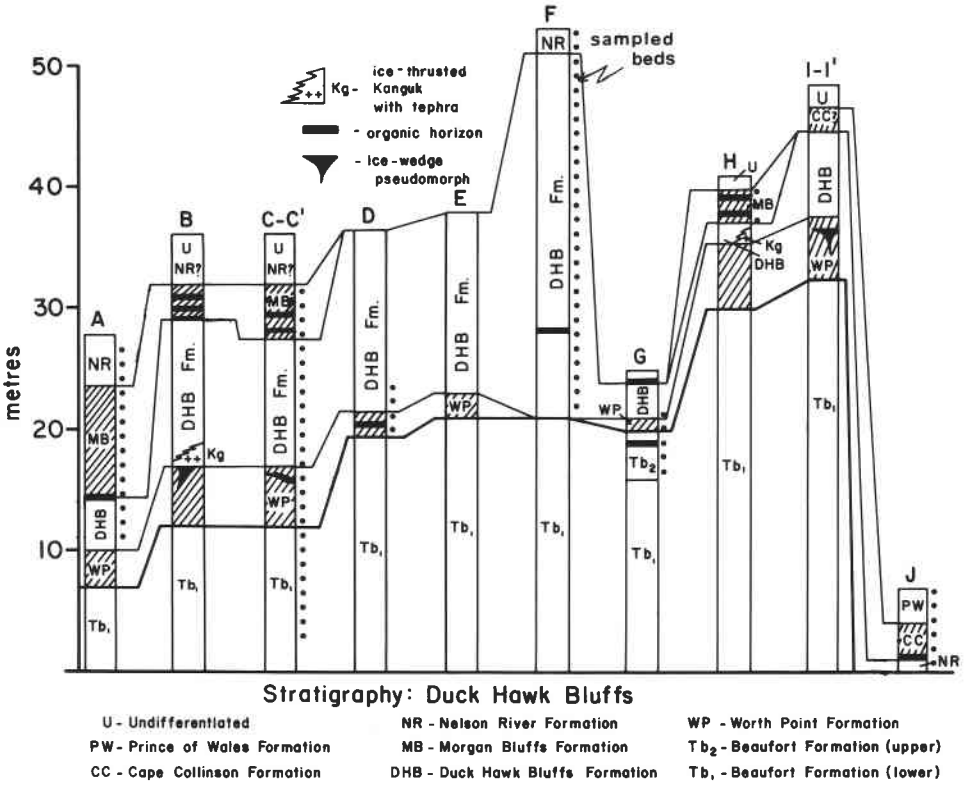


FIG. 2. Lithostratigraphy of Duck Hawk Bluffs (sections A–I) and section J (east of mouth of Mary Sachs Creek).

(1984) at Duck Hawk Bluffs, following collection of some 10 oriented samples in 1981, led to the conclusion that the Worth Point Formation and Duck Hawk Bluffs Formation sediments were magnetically reversed and therefore of Matuyama age and that the Brunhes–Matuyama boundary is probably recorded within the Morgan Bluffs Formation. Because these initial results looked promising, systematic sampling of Duck Hawk Bluffs was completed by the authors in 1983 and is reported here.

Paleomagnetism: method and results

Eleven stratigraphic units from seven sections at Duck Hawk Bluffs and adjacent bluffs on Mary Sachs Creek were sampled. In total, we obtained 386 separate and fully oriented samples by

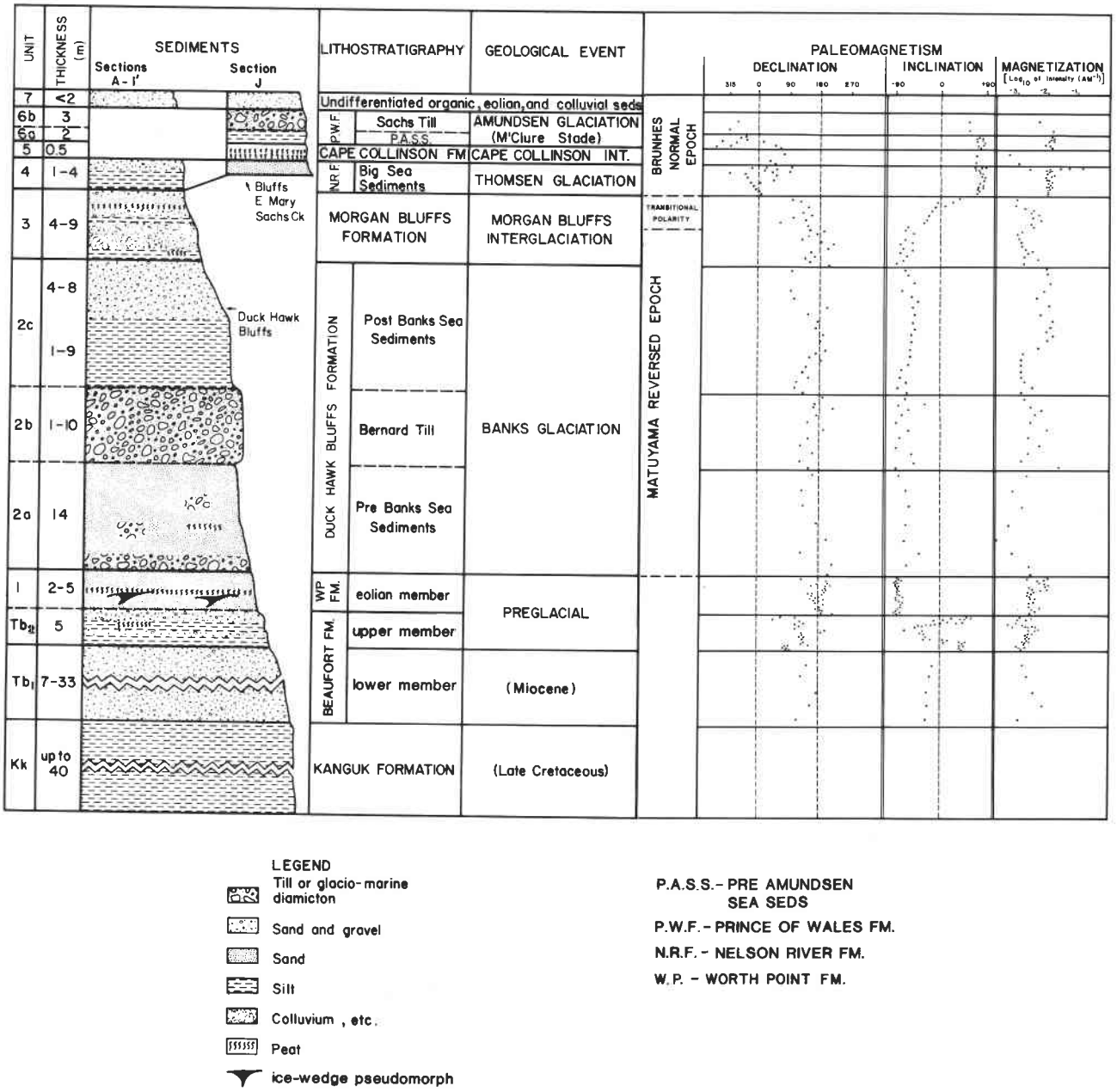


FIG. 3. Composite stratigraphic section of Duck Hawk Bluffs showing location and paleomagnetic characteristics of sampled units.

pushing plastic cubes into a vertical face that had been cleaned by shovel. The locations of the sampled units are indicated in Fig. 2. Remanence measurements were made with a Schonstedt SSM-1 magnetometer. Stepwise alternating-field demagnetization of pilot specimens was carried out with a conventional AC tumbling demagnetizer at 5, 10, 15, 20, 30, and 40 mT. Qualitative assessment of changes in magnetic vector directions and magnetization values led to a choice of 10, 30, and 50 mT as the best cleaning fields for the remaining samples. As shown in Fig. 4 and previously in Vincent *et al.* (1984), most specimens are characterized by a single-component remanent magnetization. On the basis of the low median destructive field of these specimens, it appears that detrital magnetite is the predominant remanence carrier. We derived the characteristic remanent directions of pilot specimens using a modified version of the principal-component-analysis technique of Kirschvink (1980). Modification of this program involved changes in the search routine to obtain the optimum line-fit parameters (see Schmidt 1982).

Fifty percent of the samples collected provided materials that remained physically intact and yielded well-defined, internally coherent directions ($\alpha_{95} < 15°$, Table 1). Some samples were collected from frozen or wet materials, and after transport to the laboratory some of these had to be discarded because melting and (or) drying resulted in a visible disturbance of the sediments.

The sediments carry magnetite-residing magnetizations that exhibit systematic polarity correlations between temporally equivalent units from the spatially separate stratigraphic sections. Remanence data (paleoinclinations, paleodeclinations, and magnetic intensity) are presented in Fig. 3 for a composite section of Duck Hawk Bluffs. In units Tb_2, 4, and 6a, only every other sample is shown for clarity.

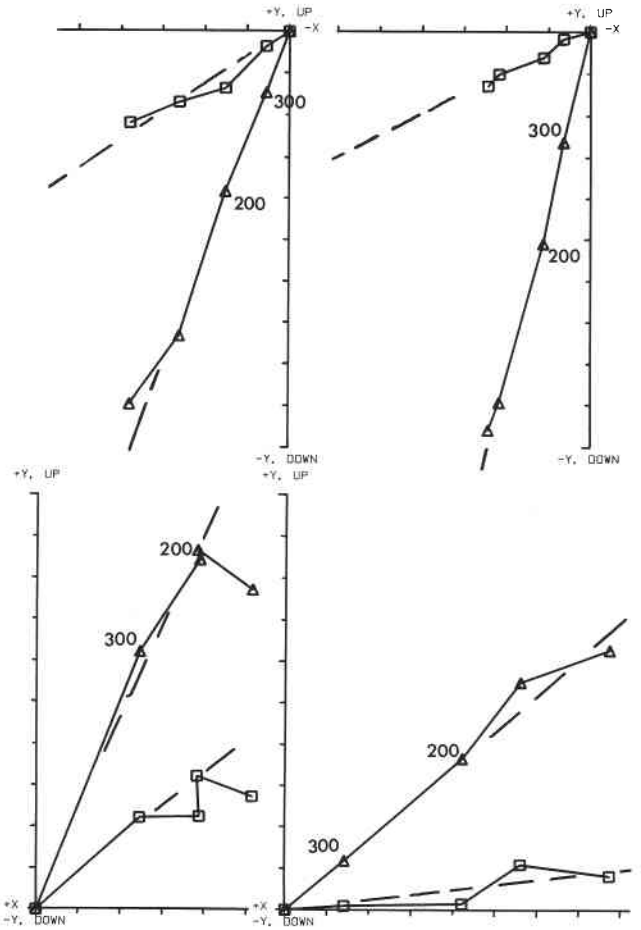


FIG. 4. Orthogonal projections (Zijderveld plots) of alternating-field-demagnetization characteristics of two normally magnetized samples (top) and two reversely magnetized samples (bottom) from Duck Hawk Bluffs. Triangles are projections on vertical plane; squares are projections on horizontal plane. North is $+Y$, east is $-X$.

In general, the Banks Island sediments are weakly magnetized, with intensities ranging from 0.001 to 0.2 A m^{-1}. The weak intensities no doubt result from the fact that the late Cenozoic sediments are mainly derived from Mesozoic and early Cenozoic beds that have a low ferromagnetic mineral content. Considering the expected magnetic inclination for this sampling latitude for an axial geocentric dipole (+81° for 72°N latitude), the inclinations are shallow. The shallowing averages 6° for the upper, normally magnetized sediments and 11° for the lower, reversely magnetized sediments (Table 2). This could be accounted for by the gradual and progressive seaward (southerly) subsidence of the deposits. Both the negative and the positive inclined vectors are smeared toward the southeast (on average S13°E). This feature was also reported in Vincent *et al.* (1984) and was thought to result from within-till shearing by glacier overriding, as its direction is antiparallel to the direction of ice flow (Nielsen 1982). The much greater sampling density in this study has firmly established the southeasterly smear of the magnetic declination in the sediments. The sense of the inclinations and that of the declinations are internally consistent within each unit. If it is assumed that the sense of the magnetic inclination provides evidence for the polarity of the magnetization of any one unit (negative inclination equivalent to reverse polarity; positive inclination equivalent to normal polarity), systematic correlations should be apparent between temporally equivalent units from the various stratigraphic sections. In fact, the more detailed sampling carried out by the current authors has in all cases revealed consistent magnetic records from the lithostratigraphically correlated units found in seven different sections.

Table 1 contains a summary of the remanence data for each unit at each section, while Table 2 shows the calculated mean pole positions and Fig. 5 shows a stereographic plot of these positions. Quite clearly, the beds fall into three categories: (*i*) an upper group, which includes the upper Morgan Bluffs Formation and younger Nelson River, Cape Collinson, and Prince of Wales formations (units 4–6, Fig. 3), all characterized by normal polarity (positive inclination and northerly mean pole positions); (*ii*) a middle group, which includes the Worth Point, Duck Hawk Bluffs, and lower Morgan Bluffs formations (units 1–3, Fig. 3), characterized by a reversed polarity (negative inclinations and southerly mean pole positions); and (*iii*) a lower group, which comprises units of mixed polarity (units Tb_1 and Tb_2, Fig. 3), showing with increasing age a normal–reversed–normal sequence. The age of this last magnetic sequence must predate the Pleistocene on the basis of the Tertiary floral and faunal record contained in the unit (Matthews *et al.* 1986). Because of the lower sampling density of this and underlying units and the likely presence of a hiatus in the sequence, it is not possible to accurately place the magnetic record within the appropriate polarity chron. Nevertheless, further work on this and other Beaufort Formation exposures could lead to the establishment of a paleomagnetic (and chronologic) record of this critical late Neogene unit. The polarity of the upper part of the Morgan Bluff Formation sampled at unit 3, section H, is normal, whereas those of the middle and lower parts of the Morgan Bluff Formation sampled at unit 3, sections A and B, are reversed (Table 1). Thus the Brunhes–Matuyama boundary, which has been assigned an age of 790 ka (Johnson 1982), is clearly recorded within the upper Morgan Bluffs Formation at Duck Hawk Bluffs.

Conclusions

The detailed paleomagnetic investigation of Duck Hawk Bluffs on Banks Island indicates that (*i*) deposits of the Worth Point Formation and those of the Banks Glaciation (Duck Hawk Bluffs Formation) are Early Pleistocene in age or older, (*ii*) the Brunhes–Matuyama boundary is clearly recorded within the interglacial Morgan Bluffs Formation, and (*iii*) a reliable paleomagnetic record can be obtained from sediments that have been subjected to permafrost conditions for extensive periods of time.

Terrestrial sediments that are magnetically reversed and of Matuyama age have also been recognized on lands bordering the Arctic Ocean — the Kap København Formation in Greenland (Abrahamsen and Marcussen 1986), the older lacustrine deposits in the Old Crow Basin of the Yukon (Pearce *et al.* 1982), and the Gubik Formation of the Alaskan Coastal Plain (Carter and Galloway 1985) — and in the Arctic Ocean basin itself (Clarke *et al.* 1980; Aksu and Mudie 1985; Witte and Kent 1988). The Banks Island paleomagnetic record therefore provides a unique opportunity for correlating major Quaternary sequences found there with those of other areas. This is particularly important at this time, since many of the older sediment suites provide critical information on conditions that characterized Arctic

TABLE 1. Remanence data by

	A					C					D					F				
Unit	N	M (A/m)	D, I (°)		k	α_{95} (°)	N	M (A/m)	D, I (°)	k	α_{95} (°)	N	M (A/m)	D, I (°)	k	α_{95} (°)	N	M (A/m)	D, I (°)	k	α_{95} (°)
6b																					
6a																					
5																					
4	4	0.180	[1]63,	82	67	11											8	0.070	[11]53, 76	46	8
3	2	0.006	[2]147,	−69	36	43	10	0.006	[4]176, −65	24	9										
2c																	21	0.014	[12]160, −63	17	8
2b	5	0.030	[3]147,	−69	17	19	5	0.008	[5]148, −46	4	42	4	0.020	[9]175, −77	8	34	3	0.002	[13]175, −55	8	46
2a							4	0.005	[6]196, −48	10	31						9	0.005	[14]167, −68	10	17
1							13	0.008	[7]146, −14	7	16	27	0.010	[10]193 −74	51	4					
Tb_2																					
Tb_1							5	0.008	[8]148, −11	10	26										

NOTES: Section and unit labels as in Figs. 2 and 3; superscript numbers represent poles in Fig. 5. *N*, number of samples used in calculating mean directions; *M*, mean magnetization at selected cleaning field; *D*, mean declination, measured from true north; *I*, mean inclination, positive downward; *k* precision parameter (Fisher 1953); α_{95}, radius of circles of 95% confidence.

[a]Values represent mean of all sections for each unit.

[b]Samples from unit 3 show normal, transitional, and reversed directions; therefore mean directions are not determined.

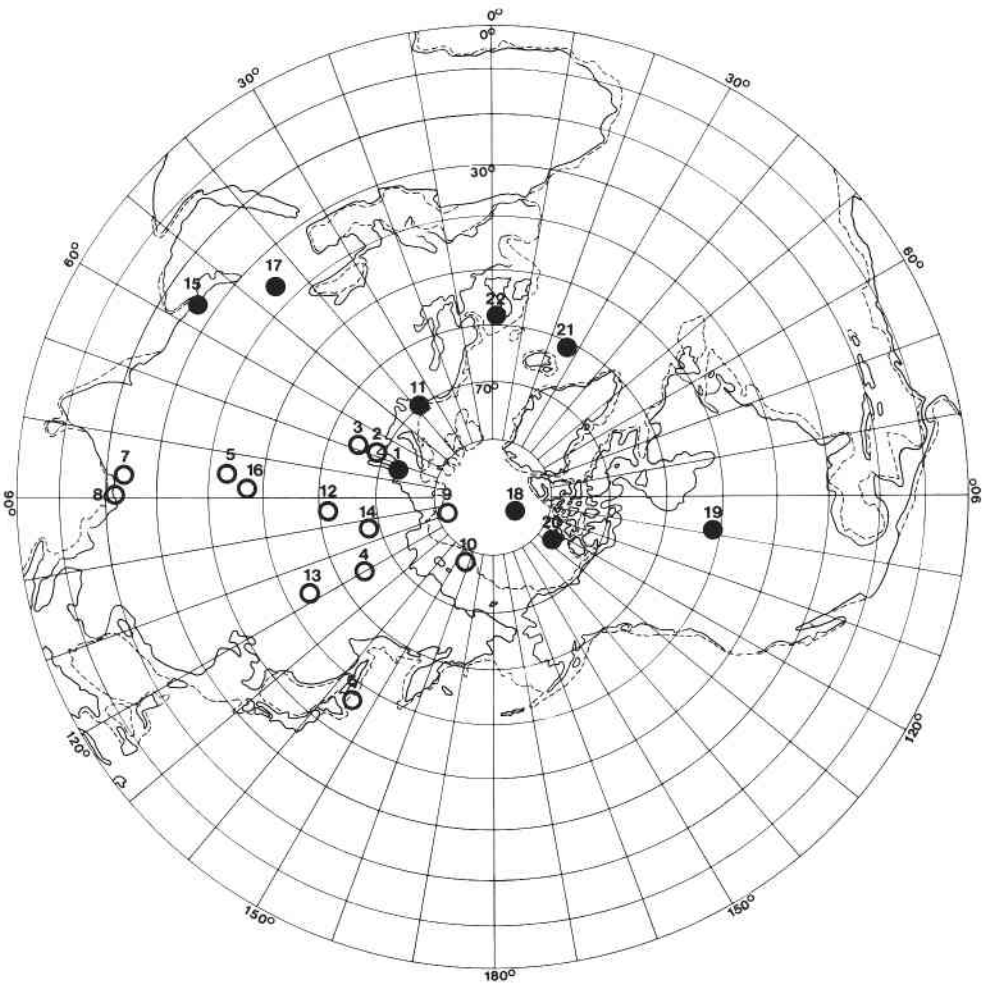


FIG. 5. Polar equal-area projection indicating mean pole positions of sampled units. Numbers correspond to superscript numbers in Table 1. Solid circles denote normal polarity; open circles denote reversed polarity.

section (A–J) and stratigraphic unit

G					H					J					Means[a]			
N	M (A/m)	D, I (°)	k	α_{95} (°)	N	M (A/m)	D, I (°)	k	α_{95} (°)	N	M (A/m)	D, I (°)	k	α_{95} (°)	N	M (A/m)	D, I (°)	
										2	0.016	[19]327, 57	67	31	1	0.016	327, 57	
										21	0.032	[20]332, 78	41	5	1	0.032	332, 78	
										4	0.033	[21]20, 63	4	50	1	0.003	20, 63	
										6	0.050	[22]39, 64	31	12	3	0.004	29, 77	
					14	0.030	[18]352, 80	225	3						—[b]		—[b]	
															1	0.014	160, −63	
															4	0.015	159, −62	
															2	0.005	186, −59	
															2	0.009	156, −47	
3	0.005	[15]88, 57	19	29											1	0.005	88, 57	
13	0.007	[16]147, −40	15	11											1	0.007	147, −40	
11	0.001	[17]84, 50	89	5											1	0.001	84, 50	
															1	0.008	148, −11	

TABLE 2. Mean poles

	N	M (A/m)	D (°)	I (°)	k	α_{95} (°)	Lat.	Long.
Mean normal	6	0.066	26	75	50	10	77°N	3°W
Mean reversed	7	0.013	165	−70	121	5.5	71°S	82°W
Mean normal and reversed	13	0.037	360	73	52	6	77°N	54°E

NOTES: Transitional polarities not included in calculations of mean pole positions. Abbreviations as in Table 1.

areas before and during the well-documented continental Quaternary glaciations. Since the Brunhes–Matuyama boundary is very seldom recognized in terrestrial sediments (the only other possible site in Canada being in the Old Crow Basin of the Yukon) (cf. Pearce *et al.* 1982), its recognition in the interglacial Morgan Bluffs Formation is critical: it provides a powerful and reliable tool for the correlation of terrestrial and marine stratigraphies (Clarke *et al.* 1984).

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